



UNIVERSITY OF  
**LEICESTER**

## **School of Computing and Mathematical Sciences**

### **CO7201 Individual Project Interim Report**

#### **A financial decision assistant web application**

**Thomas Shun Hau**

**Student email: tsh11@student.le.ac.uk**

**Student ID: 249043010**

**Project Supervisor: ZHENG, Zedong**

**Principal Marker: LIU, Daqi**

**Word Count: 836**

**27 March 2026**

## **DECLARATION**

All sentences or passages quoted in this report, or computer code of any form whatsoever used and/or submitted at any stages, which are taken from other people's work have been specifically acknowledged by clear citation of the source, specifying author, work, date and page(s). Any part of my own written work, or software coding, which is substantially based upon other people's work, is duly accompanied by clear citation of the source, specifying author, work, date and page(s). I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this module and the degree examination as a whole.

Name: Thomas Shun Hau

Date: 27 March 2026

## **Contents**

|                                      |   |
|--------------------------------------|---|
| 1. Project Requirements Update ..... | 3 |
| 2. Updated requirements list .....   | 3 |
| 3. Updated Time plan .....           | 4 |

# Project Requirements Update

After extensive discussion with the project supervisor, it is revealed that the original direction of this project, which involves the implementation of a finance decision web application and deployed using cloud services, is inappropriate for the topic and fails to demonstrate any technical skills.

A new approach of this project is to focus on implementing new functions to “assist” the user on understanding the finance information from both market data and calculation results of the web application. Possible solutions that I have researched are Natural Language Processing models (NLP) that focus on data to text translation.

Importantly, research and implantation of this new feature take time. But as of writing of this interim report, this project is approaching submission date. This means I will be using Easter holiday to catch up by researching and implementing this new function.

Also, due to time constraints, I plan to drop the containerization and cloud deployment part from this project.

Due to this change, a new requirement, time, and risk plan must be made.

## Updated Requirements list

### Essential:

Data acquisition and preprocessing

Portfolio calculation and analysis

Visualize output

Benchmark comparison

User interface

Evaluate on historic data

### Desirable:

Advanced recommendation feature

Multiple models

**Natural Language Explanation via existing model**

### Optional:

**Natural Language Explanation via self-developed model**

**Comparison of different NLP models**

**Feedback page**

## Updated Time plan

| Date                              | Tasks                                                                                                           | Explanation                                                                                                                                                                                                                                                                                                                            | Status    |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 16 Feb – 27 Feb                   | Initial research, literature review, set up development environment                                             | <ul style="list-style-type: none"> <li>- Initial research on finance concepts.</li> <li>- Reading existing work on finance algorithms.</li> <li>- Set up intellij and gitlab repo</li> </ul>                                                                                                                                           | completed |
| 2 Mar – 6 Mar                     | Begin system design                                                                                             | <ul style="list-style-type: none"> <li>- Set up basic structure of the web app, external market data API end point</li> <li>- Set up basic dashboard for single stock ticker</li> </ul>                                                                                                                                                | completed |
| 9 Mar – 13 Mar                    | Implement Algorithm                                                                                             | <ul style="list-style-type: none"> <li>- Research algorithms by reading academic papers and code examples.</li> <li>- Implement Mean-Variance Optimization</li> </ul>                                                                                                                                                                  | completed |
| 16 Mar – 20 Mar                   | Continue system implementation                                                                                  | <ul style="list-style-type: none"> <li>- Research Containerization and cloud deployment</li> <li>- Implement portfolio calculation.</li> <li>- Implement portfolio dashboard.</li> <li>- Implement visualization of portfolio finance data.</li> </ul>                                                                                 | completed |
| 23 Mar – 27 Mar<br>(this week)    | Changing project direction, and research corresponding materials                                                | <ul style="list-style-type: none"> <li>- Feedback from project supervisor revealed the need to change project direction.</li> <li>- Now researching tools to “assist” user. E.g. NLP models to translate finance data and terminology to layman terms.</li> <li>- This is done by studying existing work, papers and books.</li> </ul> | ongoing   |
| 30 Mar – 3 Apr<br>(Easter week 1) | Continue research on new direction, and begin it's implementation                                               | <ul style="list-style-type: none"> <li>- Continue studying.</li> <li>- By Wednesday 1 Apr, begin implementing knowledge and techniques learned from studying.</li> <li>- A basic task like implement existing code or model.</li> </ul>                                                                                                | Pending   |
| 6 Apr – 10 Apr<br>(Easter week 2) | Continue and complete implementation of all desirable requirements. Finish prototype for Principal Marker Demo. | <ul style="list-style-type: none"> <li>- Continue implementing the model, fine tuning to this project's needs.</li> <li>- Implement backtesting for portfolios.</li> </ul>                                                                                                                                                             | Pending   |

|                    |                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                              |         |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 13 Apr –<br>17 Apr | Principal Marker Demo. Change project based on feedback. Begin report writing and coding own implementation. Look for test participants. | <ul style="list-style-type: none"> <li>- Principal Marker Demo on Wed 15/04/2026 10:30 a.m.</li> <li>- Change project depending on feedback received.</li> <li>- At the same time, begin gathering evidence for report.</li> <li>- Begin implement self-developed model.</li> <li>- Begin looking for participant to use web app to get feedback.</li> </ul> | Pending |
| 20 Apr –<br>24 Apr | Buffer week. Continue coding and report writing.                                                                                         | <ul style="list-style-type: none"> <li>- Continue coding and organising existing work.</li> <li>- Complete self-developed model</li> <li>- Have tester try web app, and record feedback</li> <li>- Receive feedback from supervisor, changes based on feedback</li> </ul>                                                                                    | Pending |
| 27 Apr –<br>1 May  | Complete and submit final report template. Final changes to coding part.                                                                 | <ul style="list-style-type: none"> <li>- Final report template (formative)</li> <li>- Final changes to coding part</li> <li>- Implement missing tests</li> <li>- Implement missing comparison results</li> </ul>                                                                                                                                             | Pending |
| 4 May –<br>8 May   | Wrap up on project on both dissertation and coding                                                                                       | <ul style="list-style-type: none"> <li>- Same as topic</li> </ul>                                                                                                                                                                                                                                                                                            | Pending |
| 11 May –<br>15 May | Project conclusion.                                                                                                                      | <ul style="list-style-type: none"> <li>- Final report and project submission (Marked)</li> </ul>                                                                                                                                                                                                                                                             | Pending |
| 18 May –<br>22 May | Viva                                                                                                                                     | <ul style="list-style-type: none"> <li>- Viva</li> </ul>                                                                                                                                                                                                                                                                                                     | Pending |